EXERCISE 3

Peano's Counterexample: Wronskian Vanishing vs Linear Independence

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Peano Counterexample • Rigorous Analysis

PHASE 1: STATEMENT & SETUP

We have seen that, if $x_1(t)$ and $x_2(t)$ are two solutions of a linear homogeneous second order differential equation, then they are linearly independent if and only if their Wronskian is non-zero. This is not true for general functions that are not the solutions of some differential equation.

1. Check that, if $f(t) = t^2|t|$, then $\frac{df}{dt} = 3t|t|$ (straightforward for $t \neq 0$; use the formal definition of the derivative at $t = 0$).
2. Considering $f_1(t) = t^2|t|$ and $f_2(t) = t^3$, show that, although these two functions are linearly independent on $\mathbb{R}$, their Wronskian is identically zero.

PHASE 2: FUNDAMENTAL FORMULATION

Linear independence on a function space requires that $c_1 f_1(t) + c_2 f_2(t) = 0$ for all $t \in \mathbb{R}$ implies $c_1 = c_2 = 0$. The Wronskian is $W[f_1, f_2](t) = f_1(t) f_2'(t) - f_2(t) f_1'(t)$. This problem demonstrates why the condition "$W(t) \equiv 0 \implies$ linearly dependent" fails for arbitrary functions that do not satisfy a common regular linear ODE.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Part (i): Differentiating $f(t) = t^2|t|$

For $t > 0$: $|t| = t \implies f(t) = t^3 \implies f'(t) = 3t^2 = 3t(t) = 3t|t|$.

For $t < 0$: $|t| = -t \implies f(t) = -t^3 \implies f'(t) = -3t^2 = 3t(-t) = 3t|t|$.

At $t = 0$ (Formal Newton Difference Quotient):

$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{h^2|h| - 0}{h} = \lim_{h \to 0} h|h|$$

Testing left and right limits: $\lim_{h\to 0^+} h^2 = 0, \lim_{h\to 0^-}(-h^2) = 0$. Both equal 0, so $f'(0) = 0$.

Evaluating $3t|t|$ at $t = 0$: $3(0)|0| = 0$. Thus, everywhere on $\mathbb{R}$:

$$\mathbf{\frac{df}{dt} = 3t|t|}$$

Part (ii): Linear Independence of $f_1(t) = t^2|t|$ and $f_2(t) = t^3$ on $\mathbb{R}$

Assume constants $c_1, c_2 \in \mathbb{R}$ such that $c_1 f_1(t) + c_2 f_2(t) = 0$ for all $t \in \mathbb{R}$:

$$c_1 t^2|t| + c_2 t^3 = 0 \quad \forall t \in \mathbb{R}$$

Pick $t = 1$: $c_1(1)^2|1| + c_2(1)^3 = c_1 + c_2 = 0 \implies c_2 = -c_1$.

Pick $t = -1$: $c_1(-1)^2|-1| + c_2(-1)^3 = c_1(1) + c_2(-1) = c_1 - c_2 = 0 \implies c_1 - (-c_1) = 2c_1 = 0 \implies c_1 = 0,\ c_2 = 0$.

Since only the trivial combination works: **f_1 and f_2 are linearly independent on $\mathbb{R}$.**

Wronskian Determinant on $\mathbb{R}$:

$$W[f_1, f_2](t) = f_1(t)f_2'(t) - f_2(t)f_1'(t) = \left(t^2|t|\right)\left(3t^2\right) - \left(t^3\right)\left(3t|t|\right)$$

$$W[f_1, f_2](t) = 3t^4|t| - 3t^4|t| \equiv 0 \quad \forall t \in \mathbb{R}$$

PHASE 4: INTERPRETATION & EPISTEMOLOGICAL INSIGHT

WHY PEANO'S COUNTEREXAMPLE DOES NOT CONTRADICT ABEL'S THEOREM

If f_1 and f_2 were solutions of a regular second-order ODE $x'' + p_1(t)x' + p_2(t)x = 0, W \equiv 0$ would force linear dependence. However, constructing the ODE satisfied by f_1 and f_2 on $t > 0$ gives $t^2x'' - 6x = 0 \iff x'' - \frac{6}{t^2}x = 0$. The coefficient $p_2(t) = -6/t^2$ is **singular at** $t = 0$. Because the coefficient blows up at $t = 0$, the ODE is not regular, Picard-Lindelöf existence-uniqueness breaks down, and Abel's theorem does not apply across $t = 0$.